



Cyberscope

A *TAC Security* Company

Audit Report

SNVR

July 2026

Network Secret

Address `secret1d6qvapy96q94etwlnC7j33dlzmenz4j70w3ew7`

Audited by © cyberscope

Table of Contents

Table of Contents	1
Risk Classification	2
Review	3
Audit Updates	3
Source Files	3
Overview	5
Metadata	7
Findings Breakdown	9
Diagnostics	10
ST - Stops Transactions	11
Description	11
Recommendation	11
Team Update	12
SBB - Sixty-Four Bit Balance	13
Description	13
Recommendation	13
Team Update	14
FDR - Fixed Denomination Rate	15
Description	15
Recommendation	15
Team Update	16
CCR - Contract Centralization Risk	17
Description	17
Recommendation	17
Team Update	17
EUI - Experimental Untagged Implementation	18
Description	18
Recommendation	18
Team Update	18
PDPN - Possible Disabled Private Notifications	19
Description	19
Recommendation	19
Team Update	20
Summary	21
Disclaimer	22
About Cyberscope	23

Risk Classification

The criticality of findings in Cyberscope's smart contract audits is determined by evaluating multiple variables. The two primary variables are:

1. **Likelihood of Exploitation:** This considers how easily an attack can be executed, including the economic feasibility for an attacker.
2. **Impact of Exploitation:** This assesses the potential consequences of an attack, particularly in terms of the loss of funds or disruption to the contract's functionality.

Based on these variables, findings are categorized into the following severity levels:

1. **Critical:** Indicates a vulnerability that is both highly likely to be exploited and can result in significant fund loss or severe disruption. Immediate action is required to address these issues.
2. **Medium:** Refers to vulnerabilities that are either less likely to be exploited or would have a moderate impact if exploited. These issues should be addressed in due course to ensure overall contract security.
3. **Minor:** Involves vulnerabilities that are unlikely to be exploited and would have a minor impact. These findings should still be considered for resolution to maintain best practices in security.
4. **Informative:** Points out potential improvements or informational notes that do not pose an immediate risk. Addressing these can enhance the overall quality and robustness of the contract.

Severity	Likelihood / Impact of Exploitation
● Critical	Highly Likely / High Impact
● Medium	Less Likely / High Impact or Highly Likely/ Lower Impact
● Minor / Informative	Unlikely / Low to no Impact

Review

Audit Updates

Initial Audit	14 Jul 2026
---------------	-------------

Source Files

Filename	SHA256
batch.rs	659cfe6b1dcfe6cf68d9833a59364dbcc1d0e3a44add1a05a808c056bbb4d9a2
btbe.rs	b05e68a2c9df5cfb513e75cf153de2ce6f8f32fb5a27443c23fa33ebb6f34209
constants.rs	c155515d167bbccb3a19e26c2ecc5ec2ac4cc7cbd4c3ed946fb1ac201440da42
contract.rs	3186e32aeb611a2cf1501e749cb5660f870c0179747bbc360fc3e89e268744bd
dwb.rs	fe032519347a96f6f86c20bc9048cce77877fa377e99c2e911455107754dafa7
execute.rs	85e63fe17414db014f529f42133a1cb41da843e8ab41dc911d3fbac7b2c7230f
execute_admin.rs	d4d8f58cdb61c3b3281388297e0f003fb8dffeabc45f0fcc61ce0a697d8df249b
execute_deposit_redeem.rs	2b8246a55b51cde56fdaaf41a6eda7c37c1e6262f6fa2612dd465ea3a4c8bc00
execute_mint_burn.rs	222a1e9290ea17c3d94ec779c630a00dbc405ac9667a6e9ade31f049f9cdd7c5

execute_transfer_send.rs	66903a1402bddd24b06e48d932fd4a4e72ece3ada7e95b088152ed4fc79e750a
gas_tracker.rs	624159d62e68f21d4eea24675158a88c19ec468450813c8fbffb7e2fb9305492
lib.rs	90d40f219b0beec1be79ce19605a24387d479dc5ae42f8c8861461e7609fa518
msg.rs	e00e85b7a8ba539c6f8abd1a8023d834306b63c8e0bf273c717b4595a62a4a69
notifications.rs	74d84ac61041c6e5790c90e2842e795952d9aa6c809b007f6a1ac3326f827b53
query.rs	7f105bbf80e157fc5a845d0338040c99b3dcb78c542d2ee8945636d26fb560c8
receiver.rs	b4b222c3cd7b742e11f10e61c4ea24be69854fbd485d8d8545fd74214052103a
state.rs	8ed8ced444a73242fb517417099b9c6537ef0ed11f0dfd3a021d4153fa912a90
strings.rs	9bf034f337b813ec0893979905ed109f3c4d79135a6236b69987ce4d7b30e732
transaction_history.rs	e1d2d7f3d55859aeed02a7dc9284b29fc74e0a669734ccd90390c66a219f6bc0

Overview

The SNVR Token is a privacy-preserving fungible token smart contract written in Rust using the CosmWasm framework. It implements the SNIP-20 token standard together with the SNIP-24 and SNIP-24.1 query-permit extensions, the SNIP-25 advanced-privacy features, and the SNIP-52 private push-notification extension.

This report focuses on the deployment-specific risk profile of the token, including the features enabled at instantiation, the powers retained by privileged accounts, the implementation and dependency versions used for deployment, and the compatibility of the selected configuration with the token's intended supply and operational model. It does not include a full line-by-line security audit of the underlying contract logic. Accordingly, the code implementation, which is a direct copy of the official Secret Network reference implementation published at <https://github.com/scrtlabs/snip20-reference-impl>, is considered out of scope for this review.

Functionally, the contract provides the standard set of token operations. Holders can transfer tokens to other addresses, send tokens to other contracts along with a callback message, and delegate spending to third parties through allowances via transfer-from and send-from operations. Batch variants of these operations are available, allowing multiple transfers, sends, mints, or burns to be executed within a single transaction. The contract also supports minting and burning of tokens, as well as a deposit and redeem mechanism that wraps and unwraps the network's native SCRT coin at a one-to-one rate. Whether the deposit, redeem, mint, and burn capabilities are active is determined by configuration flags that are set at instantiation.

As a privacy token, account balances and transaction histories are stored in encrypted form and are not publicly visible on-chain. Access to this private data is authenticated either through viewing keys or through signed query permits, so that only the holder, or an approved party in the case of allowances, can read the corresponding information. To conceal storage-access patterns, the contract routes incoming balance updates through a Delayed Write Buffer (DWB) and stores settled balances in a Bitwise-Trie of Bucketed Entries (BTBE). It additionally emits SNIP-52 encrypted push notifications for received, spent, and allowance events, which wallets can decrypt to detect activity privately.

The contract defines a single administrator role, established at instantiation. The administrator can set the contract status level, which includes pausing token operations, transfer the administrator role to another address, manage the set of minter accounts, add or remove supported deposit denominations, and enable or disable the SNIP-52 notifications. Alongside the authenticated queries, a set of public queries exposes non-sensitive metadata such as the token name, symbol, decimals, configuration flags, exchange rate, minter list, list of notification channels, and contract status.

Metadata

The on-chain metadata confirms that the deployed contract corresponds to the SNVR source code, as its hash matches the reproducibly compiled source. The token is configured with the name and symbol `SNVR`, uses nine decimal places, and keeps its total supply private, which is why `total_supply` is returned as null. Minting, deposits, and redemptions are permanently disabled, while token burning is enabled. No supported deposit denominations or authorized minters are configured. The contract is currently operating normally under the `normal_run` status. Deployment records identify it as Code ID `2473`, created at block `24411104`, and show that no chain-level migration administrator or IBC port is assigned.

Code hash:

ff84f11b7639a1012126559dad4b41d5bf698b69657b09d7aea8483fe372c500

Token info: {

```
"token_info": {  
  "name": "SNVR",  
  "symbol": "SNVR",  
  "decimals": 9,  
  "total_supply": null  
}
```

}

Token configuration: {

```
"token_config": {  
  "public_total_supply": false,  
  "deposit_enabled": false,  
  "redeem_enabled": false,  
  "mint_enabled": false,  
  "burn_enabled": true,  
  "supported_denoms": []  
}
```

}

Minters: {

```
"minters": {  
  "minters": []  
}
```

}

Contract status: {

```
"contract_status": {  
  "status": "normal_run"  
}
```

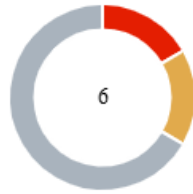
}

Deployment information: {

```
"contract_address":  
"secret1d6qvapy96q94etwlnC7j33dlzmenz4j70w3ew7",  
"contract_info": {  
  "code_id": "2473",  
  "creator": "secret16u32q8ymjw0edhkwhx6vj5hfzh6l8w5jw4sr74",  
  "label": "SNVR-1773979192463",  
  "created": {  
    "block_height": "24411104",  
    "tx_index": "0"  
  },  
  "ibc_port_id": "",  
  "admin": "",  
  "admin_proof": null,  
  "require_governance": false  
}
```

}

Findings Breakdown



●	Critical	1
●	Medium	1
●	Minor / Informative	4

Severity	Unresolved	Acknowledged	Resolved	Other
● Critical	0	1	0	0
● Medium	0	1	0	0
● Minor / Informative	0	4	0	0

Diagnostics

● Critical ● Medium ● Minor / Informative

Severity	Code	Description	Status
●	ST	Stops Transactions	Acknowledged
●	SBB	Sixty-Four Bit Balance	Acknowledged
●	FDR	Fixed Denomination Rate	Acknowledged
●	CCR	Contract Centralization Risk	Acknowledged
●	EUI	Experimental Untagged Implementation	Acknowledged
●	PDPN	Possible Disabled Private Notifications	Acknowledged

ST - Stops Transactions

Criticality	Critical
Location	src/contract.rs#L203 src/execute_admin.rs#L68
Status	Unresolved

Description

The contract administrator has the authority to stop transfers, sends, mints, burns, deposits and allowances for all users with no time limit and no user recourse. As a result, the contract may operate as a honeypot.

```
ContractStatusLevel::StopAll | ContractStatusLevel::StopAllButRedeems
=> {
    let response = match msg {
        ExecuteMsg::SetContractStatus { level, .. } => { ... }
        ExecuteMsg::Redeem { amount, denom, .. }
            if contract_status ==
ContractStatusLevel::StopAllButRedeems => { ... }
        _ => Err(StdError::generic_err(
            "This contract is stopped and this action is not allowed",
        )),
    };
    return pad_handle_result(response, RESPONSE_BLOCK_SIZE);
}
```

Recommendation

The team should carefully manage the administrator account. We strongly recommend a powerful security mechanism that will prevent a single user from accessing the contract admin functions.

Temporary Solutions:

These measurements do not decrease the severity of the finding

- Introduce a time-locker mechanism with a reasonable delay.
- Introduce a multi-signature wallet so that many addresses will confirm the action.
- Introduce a governance model where users will vote about the actions.

Permanent Solution:

- Renouncing the admin role, which will eliminate the threats but it is non-reversible.

Team Update

We acknowledge that the admin role can set contract status and pause transfers. This is part of the SNIP-20 reference design.

We will manage this operationally:

Restrict admin key access to trusted operators only

Prefer multi-signature / shared custody for the admin key where feasible

Evaluate admin renounce after launch operations are stable (irreversible)

This is not an active pause incident; it is a privileged-role risk we accept and control operationally.

SBB - Sixty-Four Bit Balance

Criticality	Medium
Location	src/dwb.rs#L319,454 src/btbe.rs#L33,194
Status	Unresolved

Description

Per-account balances are stored as eight-byte unsigned integers in both the delayed write buffer and the bucketed trie entries, so the largest representable single balance is about $1.84e19$ base units, which at the intended nine decimals corresponds to roughly 18.4 billion whole tokens per account. When an incoming amount would push an entry above this ceiling the buffer path returns an overflow error and the settle path saturates the value, meaning that a token whose supply or per-holder balance approaches this limit would suffer failing transfers or a silent capping of balances. The intended supply and any maximum expected single-account balance must therefore be kept well below this limit for the current storage layout to remain safe.

```
const DWB_AMOUNT_BYTES: usize = 8;
// ...
fn add_amount(&mut self, add_tx_amount: u128) -> StdResult<u64> {
    let mut amount = self.amount()?;
    let add_tx_amount_u64 = amount_u64(Some(add_tx_amount))?;
    safe_add_u64(&mut amount, add_tx_amount_u64);
    self.set_amount(amount)?;
    Ok(amount)
}
```

Recommendation

The team is advised to confirm that the maximum intended total supply and the largest plausible single-account balance remain far below the sixty-four bit ceiling, and to widen the amount and balance byte widths to sixteen bytes if larger figures are ever required. This will help ensure that no legitimate transfer can fail or silently lose value because of the storage representation.

Team Update

Our intended total supply and per-account balances remain well below the ~18.4B-token ceiling (at 9 decimals) implied by 64-bit storage.

FDR - Fixed Denomination Rate

Criticality	Minor / Informative
Location	src/execute_deposit_redeem.rs#L26
Status	Unresolved

Description

When multiple native denominations are configured as supported, the deposit handler sums the base-unit amounts of all provided coins and credits the total one to one without any per-asset pricing, so two different denominations with different real-world values are treated as economically equivalent when minting the wrapped token. If the deployment ever enables deposits with more than one supported denomination of differing value, a depositor could acquire the token using the cheaper asset at the same rate as the more valuable one, creating an arbitrage and accounting imbalance; the risk is inactive while deposits or multi-denomination support are disabled.

```
for coin in &info.funds {
    if constants.supported_denoms.contains(&coin.denom) {
        amount += coin.amount
    } else {
        return Err(StdError::generic_err(format!(
            "Tried to deposit an unsupported coin {}", coin.denom
        )));
    }
}
```

Recommendation

The team is advised to support only a single native denomination for deposit and redeem, or to introduce explicit per-denomination pricing if multiple assets must be accepted. This will help ensure that the wrapped token is always backed at an economically correct rate and prevents value leakage between denominations.

Team Update

Deposit functionality was disabled at instantiation (`enable_deposit: false`). Multi-denom deposit pricing risk is therefore inactive for this deployment.

CCR - Contract Centralization Risk

Criticality	Minor / Informative
Location	src/*.rs
Status	Unresolved

Description

The contract's functionality and behavior are heavily dependent on external parameters or configurations. While external configuration can offer flexibility, it also poses several centralization risks that warrant attention. Centralization risks arising from the dependence on external configuration include Single Point of Control, Vulnerability to Attacks, Operational Delays, Trust Dependencies, and Decentralization Erosion.

Recommendation

To address this finding and mitigate centralization risks, it is recommended to evaluate the feasibility of migrating critical configurations and functionality into the contract's codebase itself. This approach would reduce external dependencies and enhance the contract's self-sufficiency. It is essential to carefully weigh the trade-offs between external configuration flexibility and the risks associated with centralization.

Team Update

We acknowledge centralization risk from the admin role.

Mitigation is operational: restricted admin custody, and evaluation of multisig / eventual renounce after operations stabilize.

EUI - Experimental Untagged Implementation

Criticality	Minor / Informative
Location	src/*.rs
Status	Unresolved

Description

The contract is built from the upstream master branch of the official reference implementation at <https://github.com/scrtlabs/snip20-reference-impl>, which contains delayed-write-buffer, bucketed-trie and SNIP-52 notification functionality that is not covered by an officially released SNIP standard, and it pins the `secret-toolkit` and `secret-toolkit-crypto` dependencies to an unreleased git revision of a third-party fork.

Recommendation

The team is advised to migrate to an officially tagged release of the reference implementation from <https://github.com/scrtlabs/snip20-reference-impl/tags> and to a published release of the toolkit dependencies before or shortly after mainnet deployment, or to document and justify the specific pinned revision if the unreleased features are required.

Team Update

We acknowledge that the deployment is based on the SNIP-20 reference implementation with pinned toolkit revisions, including privacy features beyond older tagged baselines.

We accept this for current mainnet operation and can document/justify the pinned revision. Migration to an official tagged release may be considered in a future upgrade cycle.

PDPN - Possible Disabled Private Notifications

Criticality	Minor / Informative
Location	src/execute_admin.rs#L147
Status	Unresolved

Description

The administrator can call the set notification status function to disable SNIP-52 notifications globally, after which transfer, spend and allowance operations continue to function normally but no longer emit the encrypted notification payloads that wallets rely on to detect incoming activity privately. As a result the admin can degrade the privacy and usability experience for all users, causing wallets to miss real-time encrypted updates, without stopping the underlying token operations and without any per-user consent.

```
pub fn set_notification_status(deps: DepsMut, enabled: bool) ->
    StdResult<Response> {
    NOTIFICATIONS_ENABLED.save(deps.storage, &enabled)?;

    Ok(Response::new().set_data(to_binary(&ExecuteAnswer::SetNotificationStatus {
        status: Success,
    })))
}
```

Recommendation

The team should carefully manage the administrator account. We strongly recommend a powerful security mechanism that will prevent a single user from accessing the contract admin functions.

Temporary Solutions:

These measurements do not decrease the severity of the finding

- Introduce a time-locker mechanism with a reasonable delay.
- Introduce a multi-signature wallet so that many addresses will confirm the action.
- Introduce a governance model where users will vote about the actions.

Permanent Solution:

- Renouncing the admin role, which will eliminate the threats but it is non-reversible.

Team Update

We acknowledge that admin can disable SNIP-52 notifications globally.

We will treat notification status as an admin-controlled setting and will not disable it during normal operations without operational need.

Summary

SNVR contract implements a token mechanism. This audit assesses the token's deployment-specific risk profile, including its enabled features, privileged capabilities, implementation version, supply-related limitations, and relevant operational dependencies. It does not include a full line-by-line security audit of the underlying contract logic. It should be noted that smart contracts on the Secret Network are upgradable: the network provides a contract-migration mechanism through which a deployed contract's bytecode can be replaced with a new version by its designated migration administrator. The audited source was reproducibly compiled, and the resulting WASM hash was confirmed to match the code hash of the deployed contract.

Disclaimer

The information provided in this report does not constitute investment, financial or trading advice and you should not treat any of the document's content as such. This report may not be transmitted, disclosed, referred to or relied upon by any person for any purposes nor may copies be delivered to any other person other than the Company without Cyberscope's prior written consent. This report is not nor should be considered an "endorsement" or "disapproval" of any particular project or team. This report is not nor should be regarded as an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Cyberscope to perform a security assessment. This document does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors' business, business model or legal compliance. This report should not be used in any way to make decisions around investment or involvement with any particular project. This report represents an extensive assessment process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Cyberscope's position is that each company and individual are responsible for their own due diligence and continuous security. Cyberscope's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies and in no way claims any guarantee of security or functionality of the technology we agree to analyze. The assessment services provided by Cyberscope are subject to dependencies and are under continuing development. You agree that your access and/or use including but not limited to any services reports and materials will be at your sole risk on an as-is where-is and as-available basis. Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives and other unpredictable results. The services may access and depend upon multiple layers of third parties.

About Cyberscope

Cyberscope is a TAC blockchain cybersecurity company that was founded with the vision to make web3.0 a safer place for investors and developers. Since its launch, it has worked with thousands of projects and is estimated to have secured tens of millions of investors' funds.

Cyberscope is one of the leading smart contract audit firms in the crypto space and has built a high-profile network of clients and partners.



A **TAC Security** Company

The Cyberscope team

cyberscope.io